

OMNICTF - Shibliu (Shibuya)

A Minecraft world-forensics writeup, told from the solver's point of view

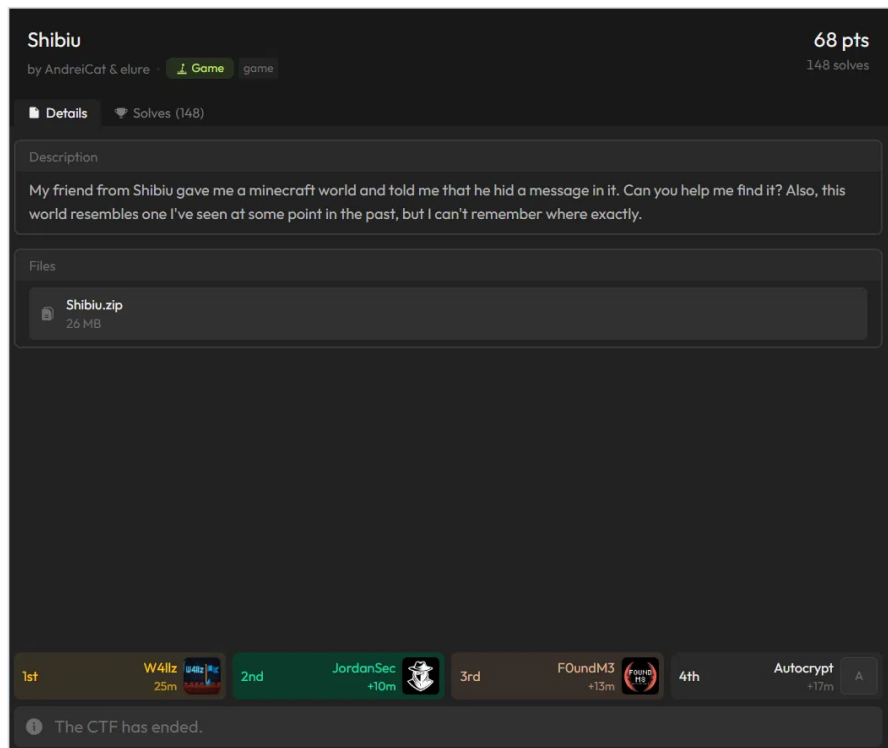
Category	Game / Forensics	Points	68
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Result	2nd blood (+10 minutes), 148 total solves		

OMNICTF{M1N3CR4FT_IZ_FUN}

The clue in the prompt

The challenge came with a single 26 MB Minecraft world and a short story: a friend from "Shibliu" had hidden a message inside it. The last sentence said the world looked familiar, which felt much more important than it first appeared.

If this was a modified public map, I did not need to search an entire city block by block. I only needed to identify the original, compare the two copies, and inspect whatever had changed.



The challenge page gave me one Minecraft save, a familiar-looking world, and 68 points to work for.

First look at the save

I started in WSL by extracting the archive.

```
(M1r4ge@brothers)-[/mnt/c/users/fayez/Downloads/ctf]  
$ unzip Shibiu.zip
```

My first step was simply extracting Shibiu.zip in WSL.

```
unzip Shibiu.zip  
cd Shibiu  
ls
```

```
(M1r4ge@brothers)-[/mnt/c/users/fayez/Downloads/ctf/Shibiu]  
$ ls  
advancements  DIM-1  icon.png  lunar_hosted_world_data.json  poi  stats  
data          DIM1   level.dat  minimap  region  
datapacks     entities level.dat_old playerdata session.lock
```

The extracted folder had the usual Java-world structure: level.dat, region, entities, and playerdata.

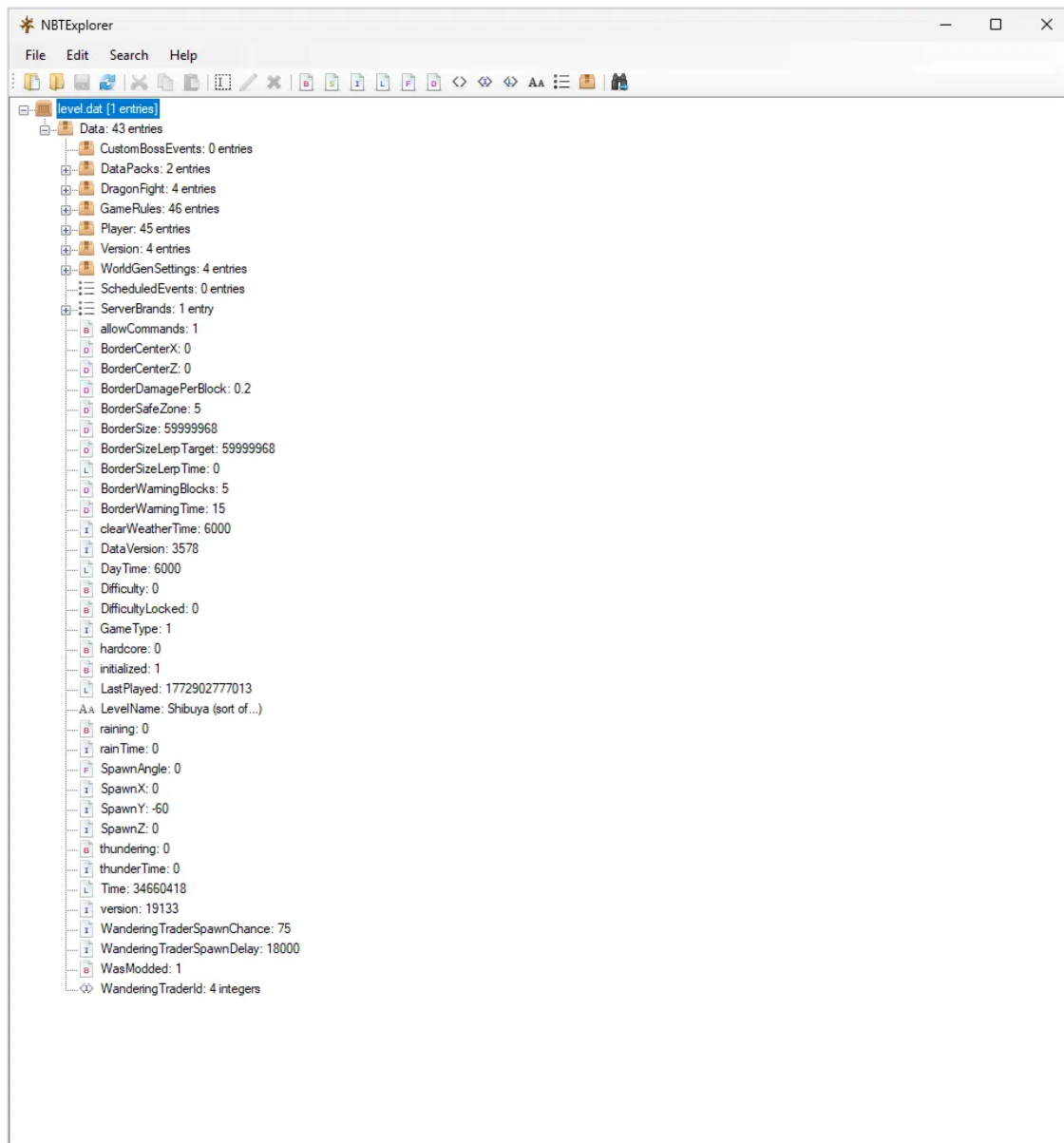
The folder looked like a normal Minecraft Java save using the Anvil format. It also contained 99 map_*.dat files. I checked them briefly, but they were ordinary overhead maps of the city rather than pieces of the flag.



Even the tiny icon.png preview looked like a city intersection.

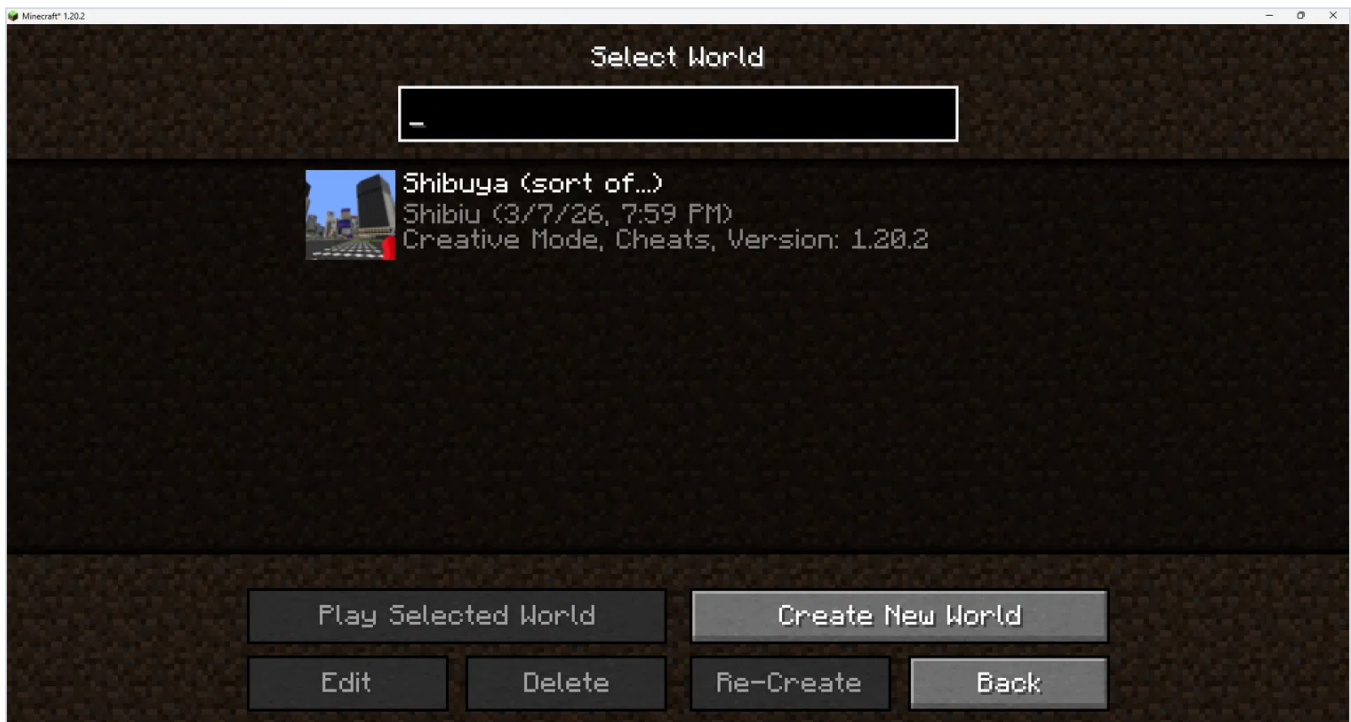
Next I opened level.dat in NBTEditor. The fields that mattered were:

- World name: Shibuya (sort of...)
- DataVersion: 3578
- Minecraft version: 1.20.2
- Spawn: (0, -60, 0)
- Game mode: Creative, with commands enabled



NBTExplorer confirmed the world name, Creative mode, DataVersion 3578, and spawn at (0, -60, 0).

Those values gave me a safe version to launch and a useful coordinate to return to later. Minecraft itself confirmed the same world name, mode, and version.



Minecraft identified the save as Shibuya (sort of...), in Creative mode on version 1.20.2.

The title now made sense: "Shibiu" was a play on "Shibuya."

Finding the clean original

The save opened onto a detailed recreation of Shibuya Crossing. I flew around for a few minutes, looking for a builder name or another searchable landmark rather than trying to inspect every building.



The challenge world was a detailed recreation of Shibuya Crossing.

Two rooftop signs stood out: NOSHY and NOSHIAGA. Searching those names together with "Minecraft" led to the builder Noshychan and a public map called **Shibuya Recreation. Sort of...**



The NOSHY rooftop sign gave me the first searchable name.



The nearby NOSHIAGA sign reinforced the same lead.

The original map was available through [MapCraft](#). At that point I had exactly what I needed:

1. the untouched public release, Shibuya_sort_of.zip;
2. the challenge copy, Shibliu.zip.

That was the turning point. Instead of searching the whole city, I could ask a much smaller question: what block states differ between these two saves?

Comparing the worlds

Minecraft region files contain compressed NBT chunks. A byte-for-byte comparison of the .mca files would be noisy because timestamps, entities, player state, and compression can change when a world is opened.

I used `compare_minecraft_worlds.py`, a small standard-library Python script that reads each chunk and hashes only its block-state sections. Matching chunks are ignored. For a differing chunk, the script decodes the palette and packed block indices, then compares every block coordinate.

My first run taught me an important lesson: compare fresh extracts. The supplied `diff.json`, made from copies that had already been opened, reported 341 blocks in 11 chunks. Two unrelated changes were mixed in: a gray stained-glass block and a chest had become air near (-14, -5, 34). That is consistent with ordinary session activity, but the artifacts alone do not say who caused it. Re-extracting both ZIPs removed the noise.

Run	Changed chunks	Changed blocks	Meaning
Opened copies (diff.json)	11	341	339 payload blocks plus 2 session changes
Fresh ZIP extracts	10	339	The actual challenge edit

The clean archive-to-archive result was:

Metric	Result
Chunks in each world	21,803
Changed chunks	10: (0,0) through (9,0)
Changed blocks	339
Every transition	minecraft:dirt -> minecraft:redstone_block
Bounds	x=0..148, y=-62, z=0..6



Rendering only the 339 changed blocks revealed a seven-pixel-high line of text at y=-62.

All 339 real changes sat on one horizontal Y layer directly below world spawn. The message extended 149 blocks along X and seven blocks along Z, so it appeared seven pixels high when viewed from above. Every changed block was dirt replaced with redstone. That pattern was far too regular to be accidental.

For a clean reproduction in WSL, I used fresh output directories and the paths from the supplied archives:

```
cd /mnt/c/Users/fayez/Downloads/ctf
mkdir -p shibiu_compare/original shibiu_compare/challenge
unzip -q Shibuya_sort_of.zip -d shibiu_compare/original
unzip -q Shibiu.zip -d shibiu_compare/challenge

python3 /mnt/c/Users/fayez/compare_minecraft_worlds.py \
"/mnt/c/Users/fayez/Downloads/ctf/shibiu_compare/original/Shibuya (sort of___)" \
"/mnt/c/Users/fayez/Downloads/ctf/shibiu_compare/challenge/Shibiu" \
--json /mnt/c/Users/fayez/diff-clean.json --ascii
```

Exposing the message

I switched to a disposable copy of the challenge world and teleported back to its recorded spawn:

```
/tp @s 0 -60 0
```

My first instinct was to clear a huge area above the flag in one command. Vanilla rejected it because I had asked it to modify 15,240,000 blocks at once. That was a useful reminder to use the exact bounds I had just recovered instead of deleting half the city.



Vanilla rejected my oversized first /fill attempt: 15,240,000 blocks was far above the 32,768-block limit.

The changed strip occupied only $x=0..148$ and $z=0..6$. On a throwaway copy, I raised the command limit just enough and cleared the narrow column above it:

```
/gamerule commandModificationBlockLimit 400000  
/fill 0 -61 0 148 319 6 air
```

Looking down at $y=-62$, the redstone blocks formed a seven-pixel font. Reading from west to east gave:

```
OMNICTF{M1N3CR4FT_IZ_FUN}
```



The exposed redstone-block message in the disposable challenge-world copy.

Closing thoughts

The Minecraft part was only half of the challenge. The decisive clue was that the city looked familiar. Once I found the public original, the solve became a clean piece of differential forensics: ignore changing metadata, compare block states, and inspect the tiny set of real edits.

The small mistake with the opened copies was also useful. It showed why forensic comparisons should begin from untouched inputs and why the transition counts matter more than a single headline number.

The flag was submitted for 68 points as the second solve overall, ten minutes behind first blood.

Tools

I used Minecraft Java 1.20.2, NBTEditor, Python 3 with `compare_minecraft_worlds.py`, the supplied `diff.json`, and MapCraft for the OSINT lead.

Flag: OMNICTF{M1N3CR4FT_IZ_FUN}